JAVA Talk 3 Mumford-Tate groups

Further reading:

[BM 99] Ben Moonen, Notes on Mumford-Tate groups.

Notation: For a G-set X and a subset $Y \subset X$, define

$$Y^G := Y \cap X^G = \{y \in Y \mid g(y) = y\}$$

$$\text{e.g. } H^{2p}(A;\mathbb{Q})^{h_{\mathbb{R}}(S^1)} = H^{2p}(A;\mathbb{Q}) \cap H^{p,p} = B^p(A)$$

1. three definitions
2. relation with Hodge cycle & Endomorphisms
3. examples.

Mumford-Tate gp

$$\langle \mathbb{V} \rangle_{\subseteq HS_{\mathbb{Q}}} = \mathrm{Rep}(MT(\mathbb{V}))$$

$$MT(\mathbb{V}) = \langle h_{\mathbb{R}}(\mathbb{C}^\times) \rangle_{\mathbb{Q}\text{-alg gp}}$$

$$= \left\{ g \in GL(\mathbb{V}) \,\middle|\, \begin{array}{l} g\alpha = \alpha \\ \forall \alpha \in Hdg^{\cdot\cdot}(\mathbb{V}) \end{array} \right\}$$

$$h_{\mathbb{R}} : \mathbb{C}^\times \longrightarrow GL(\mathbb{V}_{\mathbb{R}})$$

Tannakian monodromy gp

$$\langle V \rangle_{\subseteq LS_F(X)} = \mathrm{Rep}(\mathrm{Mon}(\rho))$$

$$\mathrm{Mon}(\rho) = \langle \rho(\pi_1(X)) \rangle_{F\text{-alg gp}}$$

$$= \left\{ g \in GL(V) \,\middle|\, \begin{array}{l} g\alpha = \alpha \\ \forall \alpha \in (T^{\cdot\cdot}(V))^{\rho(\pi_1(X))} \end{array} \right\}$$

$$\rho : \pi_1(X) \longrightarrow GL_F(V)$$

Rmk. [Toen 04, Thm 5.1.1]

schemetic homotopy type

$\forall X/\mathbb{C}$ sm proj, $\exists\, h(X) \in SHT/\mathbb{Q}$ s.t.

$$\{LS \text{ on } h(X), \otimes\} \cong {}_{\vee}MHS(X)$$

e.p. $$\{LS \text{ on } h(*), \otimes\} \cong HS_{\mathbb{Q}}$$

Q: How can we compare the following categories and compute π_1?

$${}_{\vee}MHS(X) = \mathrm{Rep}(\pi_1^{\vee MHS}(X))$$

$$LS_F(X) = \mathrm{Rep}(\pi_1^{LS}(X))$$

$$VB^{\nabla}(X) = \mathrm{Rep}(\pi_1^{dR}(X))$$

$$MHS(X) = \mathrm{Rep}(\pi_1^{hodge}(X))$$

This page is partially AI-generated; take it with a grain of salt.

Ref: [Ara 09]: Donu Arapura, arXiv: 0902.4252v2

[BZ 13]: Brosnan & El Zein, arXiv: 1302.5814v1

[HZ 87]: Hain & Zucker, unipotent variations of mix Hodge structure

Notation. [Ara09, Ex 2.2, 2.4, p13]

$$\mathcal{MHS}(X) = \{\text{admissible vMHSs on } X\}$$

$$\mathcal{UMHS}(X) = \{\text{unipotent adm vMHSs on } X\}$$

$$\mathcal{U}_r\mathcal{MHS}(X) = \left\{ \mathcal{V} \in \mathcal{UMHS}(X) \text{ with index of unipotency at most } r \right\}$$

Prop. [HZ87, (1.6)] [Ara09, Thm 5.2]

$$\mathcal{U}_r\mathcal{MHS}(X) \cong \mathrm{Rep}_{\mathcal{MHS}_{\mathbb{Q}}}\left(\mathbb{Q}[\pi_1(X,x)]/J^{r+1}\right)$$

i.e., $\rho: \mathbb{Q}[\pi_1(X,x)]/J^{r+1} \to \mathrm{End}(V)$ is a morphism in $\mathcal{MHS}_{\mathbb{Q}}$.

Among the categories, we have

$$\begin{tikzcd} & \mathcal{LS}_{\mathbb{Q}}(X) \\ \mathcal{MHS}(X) \arrow[ur] \arrow[dr] & \\ & \mathcal{VB}^{\nabla}(X) \end{tikzcd} \qquad \begin{tikzcd} & \mathcal{V}_{\mathbb{Q}} \\ (\mathcal{V}_{\mathbb{Q}}, W_\bullet, F^\bullet, \nabla) \arrow[ur, mapsto] \arrow[dr, mapsto] & \\ & (\mathcal{V}_{\mathbb{C}}, \nabla) \end{tikzcd}$$

For $\mathbb{V} \in \mathcal{vMHS}(X)$, Claude claims that

$$\mathcal{MT}(\mathbb{V}) = \langle \underbrace{h_{\mathbb{R},\eta}(\mathbb{C}^\times)}_{\text{Hodge fil}}, \underbrace{w(\mathbb{G}_m)}_{\text{wt fil}}, \underbrace{\rho(\pi_1(X,x_0))}_{\text{monodromy}} \rangle_{\mathbb{Q}\text{-alg gp}}$$

Notation: $\mathbb{V}$: $\mathbb{Q}$-Hodge structure of wt $k_{\mathbb{V}}$.

In this talk, most of time $\mathbb{V} = H^1(A;\mathbb{Q})$, $k_{\mathbb{V}} = 1$.

Rmk. We can use the Deligne torus

$$\mathbb{S} := \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}}$$

$$\begin{array}{ccccc} \mathbb{S}(\mathbb{R}) & \cong & \mathbb{C}^\times & & z \\ \downarrow & & \downarrow & & \downarrow \\ \mathbb{S}(\mathbb{C}) & \cong & \mathbb{C}^\times \times \mathbb{C}^\times & & (z,\bar{z}) \end{array}$$

to describe a (real) Hodge structure:

$$\mathbb{V} \longleftrightarrow h: \mathbb{S} \longrightarrow GL(\mathbb{V}_{\mathbb{R}}) = GL_{2n,\mathbb{R}}$$

s.t. $h_{\mathbb{R}}: \mathbb{C}^\times \longrightarrow GL(H^1(A;\mathbb{R}))$

$$a+bi \longmapsto a\,\mathrm{Id} + bJ$$

$$\begin{tikzcd} \mathbb{C}^\times \arrow[r, "h_{\mathbb{R}}"] \arrow[d, hook] & GL_{\mathbb{R}}(H^1(A;\mathbb{R})) \arrow[d, hook] \\ \mathbb{C}^\times \times \mathbb{C}^\times \arrow[r, "h_{\mathbb{C}}"] & GL_{\mathbb{C}}(H^1(A;\mathbb{C})) \end{tikzcd}$$

$$a+bi \longmapsto a\,\mathrm{Id} + bJ$$

$$\begin{array}{ccc} z & & A \\ \downarrow & & \downarrow \\ (z,\bar{z}) & & A \otimes_{\mathbb{R}} \mathbb{C} \end{array}$$

$$(z_1, z_2) \longmapsto \begin{pmatrix} z_1 1_n & \\ & z_2 1_n \end{pmatrix}$$

w.r.t. basis $\{e_1, \dots, e_n, \bar{e}_1, \dots, \bar{e}_n\}$

1. three definitions

<u>Def 1</u> $\langle \mathbb{V} \rangle_{\otimes} = \mathrm{Rep}(MT(\mathbb{V}))$

Cor. $\mathbb{W} \subseteq \mathbb{V}$ is a sub-HS iff $\mathbb{W}$ is a sub $MT(\mathbb{V})$-module.

Rmk [BM99, (1.16)]

1. $PHS_{\mathbb{Q}} \subseteq HS_{\mathbb{Q}}$ is semisimple. $\| \; W \subseteq V \rightsquigarrow W^{\perp} \subseteq V$
2. If $\mathbb{V} \in PHS_{\mathbb{Q}}$, then $MT(\mathbb{V})$ & $SMT(\mathbb{V})$ are reductive.

<u>Def 2</u> The MT gp of $\mathbb{V} \in HS_{\mathbb{Q}}$ is the smallest $\mathbb{Q}$-alg gp $G \subseteq GL(\mathbb{V})$ s.t.

$$\begin{tikzcd} \mathbb{S} \arrow[r, "h"] \arrow[dr, dashed, "\exists"'] & GL(\mathbb{V}_{\mathbb{R}}) \\ & G_{\mathbb{R}} \arrow[u, phantom, "\cup\!/"] \end{tikzcd} \qquad \begin{tikzcd} \mathbb{C}^{\times} \arrow[r, "h_{\mathbb{R}}"] \arrow[dr, dashed, "\exists"'] & GL(\mathbb{V}_{\mathbb{R}}) \\ & G(\mathbb{R}) \arrow[u, phantom, "\cup\!/"] \end{tikzcd}$$

[BM99, (1.4)] [vG94, 6.4]

i.e., $MT(\mathbb{V}) = \langle h_{\mathbb{R}}(\mathbb{C}^{\times}) \rangle_{\mathbb{Q}\text{-alg gp}}$

Similarly, the special Mumford-Tate gp / Hodge gp is

$$SMT(\mathbb{V}) = \langle h_{\mathbb{R}}(S^1) \rangle_{\mathbb{Q}\text{-alg gp}}$$

<u>Def 3</u> $MT(\mathbb{V})(\mathbb{Q}) = \{ g \in GL(\mathbb{V}) \mid g\alpha = \alpha \quad \forall \alpha \in Hdg^{\cdot\cdot}(\mathbb{V}) \}$

where

$$T^{k,l}(\mathbb{V}) = \mathbb{V}^{\otimes k} \otimes \mathbb{V}^{*,\otimes l} \qquad \text{not } (k,l)\text{-form}$$

$$Hdg^{k,l}(\mathbb{V}) = (T^{k,l}(\mathbb{V}))^{0,0} \qquad (0,0)\text{-form part of } T^{k,l}(\mathbb{V})$$

$$= \cancel{\{\alpha \in} \; (T^{k,l}(\mathbb{V}))^{h_{\mathbb{R}}(\mathbb{C}^\times)}$$

Ex. 1. Show that they are equivalent.

$$MT^2(\mathbb{V}) \begin{array}{l} \subseteq MT^1(\mathbb{V}) \\ \subseteq MT^3(\mathbb{V}) \end{array} \qquad \text{is clear.}$$

2. Give three definitions of $SMT(\mathbb{V})$.

Is the following true?

$$SMT(\mathbb{V})(\mathbb{Q}) = \{ g \in GL(\mathbb{V}) \mid g\alpha = \alpha \quad \forall \alpha \in (T^{k,l}(\mathbb{V}))^{h_{\mathbb{R}}(S^1)} \}$$

$$SMT(\mathbb{V}) = MT(\mathbb{V}) \cap SL(\mathbb{V})$$

3. Show that for $\mathbb{V} \in HS_{\mathbb{Q}}$, $\qquad (\mathbb{V}^{p,p} := \mathbb{V} \cap \mathbb{V}_{\mathbb{C}}^{p,p})$

$$\mathbb{V}^{h_{\mathbb{R}}(\mathbb{C}^\times)} = \mathbb{V}^{0,0} \qquad \mathbb{V}^{h_{\mathbb{R}}(S^1)} = \bigoplus_{p \in \mathbb{Z}} \mathbb{V}^{p,p}$$

4. Claude claims that:

For $G = MT(\mathbb{V})$, $G(\mathbb{Q})$ is Zariski dense in G, and

$$V^G = V^{G(\mathbb{Q})} \qquad Rep_{\mathbb{Q}}(G) \cong Rep_{\mathbb{Q}}(G(\mathbb{Q}))$$

Is this true? Why?

2. Relation with Hodge cycle & Endomorphisms

Rmk. By definition,

$$B^p(X) = H^{2p}(X;\mathbb{Q}) \cap H^{p,p} = H^{2p}(X;\mathbb{Q})^{h_{\mathbb{R}}(S^1)}$$

Prop. $$B^p(A) = H^{2p}(A;\mathbb{Q})^{SMT(\mathbb{V})(\mathbb{Q})}$$

Proof. $$H^{2p}(A;\mathbb{Q})^{h_{\mathbb{R}}(S^1)} \supseteq H^{2p}(A;\mathbb{Q})^{SMT(\mathbb{V})(\mathbb{R})} = H^{2p}(A;\mathbb{Q})^{SMT(\mathbb{V})(\mathbb{Q})}$$

$$h_{\mathbb{R}} : \mathbb{C}^\times \longrightarrow GL(H^1(A;\mathbb{R})) \xrightarrow{\rho_{2p,\mathbb{R}}} GL(H^{2p}(A;\mathbb{R}))$$

$$P \longmapsto \begin{pmatrix} Id_{B^p(A)} & * \\ 0 & * \end{pmatrix}$$

under some $\mathbb{Q}$-basis of $H^{2p}(A;\mathbb{Q})$

$$P := \rho_{2p,\mathbb{Q}}^{-1}\begin{pmatrix} Id_{B^p(A)} & * \\ 0 & * \end{pmatrix} \subseteq GL(H^1(A;\mathbb{Q}))$$

By def. $$B^p(A) \subseteq H^{2p}(X;\mathbb{Q})^{P(\mathbb{Q})}$$

$$\left.\begin{array}{l} P \text{ is alg over } \mathbb{Q} \\ P(\mathbb{R}) \supset h(S^1) \end{array}\right\} \Rightarrow SMT(\mathbb{V}) \subset P$$

Therefore,

$$B^p(A) \subseteq H^{2p}(X;\mathbb{Q})^{P(\mathbb{Q})} \subseteq H^{2p}(X;\mathbb{Q})^{SMT(\mathbb{V})(\mathbb{Q})} \qquad \square$$

Prop. [BM 99, (1.2.1)]

$$\begin{aligned} End(A)_{\mathbb{Q}} &= End_{HS_{\mathbb{Q}}}(\mathbb{V}) \\ &= Hdg^{1,1}(\mathbb{V}) \\ &= End_{\mathbb{Q}}(\mathbb{V})^{MT(\mathbb{V})} \\ &= End_{\mathbb{Q}}(\mathbb{V})^{SMT(\mathbb{V})} \end{aligned}$$

3. examples.

Goal. Understand the following table.

A	$MT(\mathbb{V})$	$MT(\mathbb{V})(\mathbb{C})$	$Hdg(\mathbb{V}) = SMT(\mathbb{V})$	$SMT(\mathbb{V})(\mathbb{C})$
very general AV	$GSp(\mathbb{V},\omega)$	$GSp^D(2n;\mathbb{C})$	$Sp(\mathbb{V},\omega)$	$Sp^D(2n;\mathbb{C})$
very general AV of Weil type	$GU_K(\mathbb{V},\omega) = GU_H$	$0 \to GL(n;\mathbb{C}) \to \cdots \to \mathbb{C}^\times \to 0$	$SU_K(\mathbb{V},\omega) = SU_H$	$SL(n;\mathbb{C}) = Sp^D(2n;\mathbb{C}) \cap SL_K(2n;\mathbb{C})$
EC without CM	$GL_{2,\mathbb{Q}} = GSp(\mathbb{V},\omega)$	$GL_2(2;\mathbb{C})$	$SL_{2,\mathbb{Q}} = Sp(\mathbb{V},\omega)$	$SL(2;\mathbb{C})$
EC with CM	$GU_K = GU(1,K/\mathbb{Q}) = \mathrm{Res}_{K/\mathbb{Q}}\mathbb{G}_m$	$\mathbb{C}^\times \times \mathbb{C}^\times$	$U_K = U(1,K/\mathbb{Q})$	$\mathbb{C}^\times$
$\mathbb{V} = \mathbb{Q}$	$\{Id\}$	$\{Id\}$	$\{Id\}$	$\{Id\}$
$\mathbb{V} = \mathbb{Q}(n), n \neq 0$	$\mathbb{G}_{m,\mathbb{Q}}$	$\mathbb{C}^\times$	$\{Id\}$	$\{Id\}$
[BM99, (1.21), (2.7)] $\mathbb{D} = \mathrm{End}(A)_{\mathbb{Q}}$ e.p. when $n=2$	$\subseteq GU_{\mathbb{D}}(\mathbb{V},\omega)$		$\subseteq U_{\mathbb{D}}(\mathbb{V},\omega)$ $Sp_{\mathbb{D}}(\mathbb{V},\omega)$	

Goal: Finding the equations satisfied by $a 1_{2n} + b J_\lambda$, which gives an upper bound of $MT(\mathbb{V})$.

AV case: $\omega(ix, iy) = \omega(x, y)$

$$\Leftrightarrow \quad J_\lambda^T J_D J_\lambda = J_D$$

$$\Rightarrow \quad (a 1_{2n} + b J_\lambda)^T J_D (a 1_{2n} + b J_\lambda)^T = (a^2 + b^2) J_D$$

Define

$$GSp(\mathbb{V}, \omega) = \left\{ g \in GL(\mathbb{V}) \;\middle|\; \begin{array}{l} \exists\, \nu(g) \in \mathbb{Q}^\times \text{ s.t.} \\ \omega(gx, gy) = \nu(g) \cdot \omega(x, y) \quad \forall x, y \in \mathbb{V} \end{array} \right\}$$

Then $\exists$ SES

$$0 \longrightarrow Sp(\mathbb{V}, \omega) \longrightarrow GSp(\mathbb{V}, \omega) \xrightarrow{\nu} \mathbb{G}_{m,\mathbb{Q}} \longrightarrow 0$$

and

$$MT(\mathbb{V}) \subseteq GSp(\mathbb{V}, \omega) \qquad SMT(\mathbb{V}) \subseteq Sp(\mathbb{V}, \omega)$$

K-variety case:

$$e_0 : T_0 A \longrightarrow T_0 A \quad \text{is } \mathbb{C}\text{-linear}$$

$$\Rightarrow e_{0,\mathbb{R}} : H_1(A; \mathbb{R}) \longrightarrow H_1(A; \mathbb{R}) \quad \text{is } J\text{-linear}$$

$$\Leftrightarrow \quad J_u (a 1_{2n} + b J_\lambda) = (a 1_{2n} + b J_\lambda) J_u$$

$$\Leftrightarrow \quad a 1_{2n} + b J_\lambda \in GL_K(2n; \mathbb{C})$$

Define

$$GU_K(\mathbb{V}, \omega) = GSp(\mathbb{V}, \omega) \cap GL_K(\mathbb{V}) \qquad U_K(\mathbb{V}, \omega) = Sp(\mathbb{V}, \omega) \cap GL_K(\mathbb{V})$$

$$GU^D(n, K/\mathbb{Q})(\mathbb{C}) = GSp^D(2n; \mathbb{C}) \cap GL_K(2n; \mathbb{C}) \qquad U^D(n, K/\mathbb{Q})(\mathbb{C}) = Sp^D(2n; \mathbb{C}) \cap GL_K(2n; \mathbb{C})$$

$$= \left\{ \begin{pmatrix} \alpha & -\bar{\alpha}\gamma \\ \gamma & \bar{\alpha} \end{pmatrix} \;\middle|\; \begin{array}{l} \exists\, \nu(g) \in \mathbb{C}^\times \text{ s.t.} \\ (\alpha^T - \sqrt{-1}\,\bar{\alpha}\gamma^T) D (\alpha + \sqrt{-1}\,\bar{\alpha}\gamma) = \nu(g) D \end{array} \right\} \cong GL(n; \mathbb{C})$$

Then $\exists$ SES

$$0 \longrightarrow U_K(\mathbb{V}, \omega) \longrightarrow GU_K(\mathbb{V}, \omega) \xrightarrow{\nu} \mathbb{G}_{m,\mathbb{Q}} \longrightarrow 0$$

and

$$MT(\mathbb{V}) \subseteq GU_K(\mathbb{V}, \omega) \qquad SMT(\mathbb{V}) \subseteq U_K(\mathbb{V}, \omega)$$

Weil type case: $(m = n_+ = n_-)$

We have a morphism of $\mathbb{Q}$-alg gp

$$\det_{K_{e_0}} : GL_K(\mathbb{V}) \longrightarrow \mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_{m,K}$$

Reality check:

$$\det_{K_{e_0}}(\mathbb{Q}) : GL(n; K) \longrightarrow K^\times$$

$$\det_{K_{e_0}}(\mathbb{R}) : \underbrace{GL(n;\mathbb{C})}_{= GL_K(2n;\mathbb{R})} \longrightarrow \mathbb{C}^\times$$

$$J \longmapsto i_{e_0}^{n_+}(-i_{e_0})^{n_-} = 1$$

$$a 1_{2n} + bJ \longmapsto (a+bi_{e_0})^{n_+}(a-bi_{e_0})^{n_-} = (a^2+b^2)^m$$

Rmk. We use the iso

$$\begin{array}{ccccccc} \mathbb{V} \otimes_{\mathbb{Q}} \mathbb{R} & \hookrightarrow & H^1(A;\mathbb{C}) & \twoheadrightarrow & T_0^* A & \cong & V_+ \oplus V_- \\ \circlearrowleft J & & & & \circlearrowleft i & & \uparrow i_{e_0} = i \quad \uparrow i_{e_0} = -i \end{array}$$

Define $SL_K(\mathbb{V}) := \ker \det_{K_{e_0}}$, we get a SES

$$0 \longrightarrow SL_K(\mathbb{V}) \longrightarrow GL_K(\mathbb{V}) \xrightarrow{\det_{K_{e_0}}} \mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_{m,K} \longrightarrow 0$$

Write

$$\begin{aligned} SU_K(\mathbb{V},\omega) &:= Sp(\mathbb{V},\omega) \cap SL_K(\mathbb{V}) \\ &= U_K(\mathbb{V},\omega) \cap SL_K(\mathbb{V}) \end{aligned}$$

we see that

$$SMT(\mathbb{V}) \subseteq SU_K(\mathbb{V},\omega).$$

EC case. We can compute J_λ explicitly,
but finding an equation is still hard.

$$\mathcal{T} = \begin{pmatrix} \tau & 1 \\ \bar{\tau} & 1 \end{pmatrix} \quad \mathcal{T}^{-1} = \frac{1}{2i\,\mathrm{Im}\,\tau}\begin{pmatrix} 1 & -1 \\ -\bar{\tau} & \tau \end{pmatrix}$$

$$J_\lambda = \mathcal{T}^{-1} J_e \mathcal{T} = \frac{1}{\mathrm{Im}\,\tau}\begin{pmatrix} \mathrm{Re}\,\tau & 1 \\ -|\tau|^2 & -\mathrm{Re}\,\tau \end{pmatrix}$$

Ex. check the case $\tau = i$ by hand.

When $\mathbb{Q}(\tau)$ is imaginary quadratic field (i.e., $[\mathbb{Q}(\tau):\mathbb{Q}]=2$)

$\exists\ a_1, a_2 \in \mathbb{Q}$ s.t.

$$\tau^2 + a_1\tau + a_2 = 0.$$

Therefore,

$$\tau\,(\tau \quad 1) = (\tau \quad 1)\,J_\tau \qquad J_\tau = \begin{pmatrix} -a_1 & 1 \\ -a_2 & 0 \end{pmatrix} \in M^{2\times 2}(\mathbb{Q})$$

$$\begin{pmatrix} \tau & \\ & \bar{\tau} \end{pmatrix}\begin{pmatrix} \tau & 1 \\ \bar{\tau} & 1 \end{pmatrix} = \begin{pmatrix} \tau & 1 \\ \bar{\tau} & 1 \end{pmatrix} J_\tau$$

$$\rightsquigarrow \qquad J_\tau = \mathcal{T}^{-1}\begin{pmatrix} \tau & \\ & \bar{\tau} \end{pmatrix}\mathcal{T}$$

$$\rightsquigarrow \qquad J_\tau J_\lambda = J_\lambda J_\tau$$

Define

$$GU_\tau(\mathbb{Q}) := \{ g \in GL_2(\mathbb{Q}) \mid J_\tau g = g J_\tau \}$$
$$U_\tau(\mathbb{Q}) := \{ g \in SL_2(\mathbb{Q}) \mid J_\tau g = g J_\tau \}$$

Then

$$MT(\mathbb{V}) \subseteq GU_\tau \qquad SMT(\mathbb{V}) \subseteq U_\tau$$

Rmk.

$$\det(a\,1_2 + b\,J_\lambda) = \det(a\,1_2 + b\,J_e) = (a+bi)(a-bi) = a^2+b^2.$$

Opposite direction:

Since $J_\lambda \in Lie(SMT(\mathbb{V})(\mathbb{C}))$, this gives a lower bound of $SMT(\mathbb{V})$.

For specific J_λ, it is still painful to verify if

$$\langle J_\lambda \rangle = Lie(SMT(\mathbb{V})(\mathbb{C})),$$

but if we are working on very general elements in a family, then

$$\langle \underbrace{ad(g) J_{\lambda_0}}_{= g J_{\lambda_0} g^{-1}} \rangle_{g \in G} \subseteq Lie(SMT(\mathbb{V})(\mathbb{C}))$$

$\uparrow$ gp of the period domain

$$\Rightarrow \langle X J_{\lambda_0} - J_{\lambda_0} X \rangle_{X \in Lie\, G} \subseteq Lie(SMT(\mathbb{V})(\mathbb{C}))$$

we can then choose a special J_{λ_0} in the family and compute it.

Rmk. For $\mathbb{W} \in \langle \mathbb{V} \rangle_{\otimes}$,

$$\mathbb{W}^{SMT(\mathbb{V})(\mathbb{Q})} \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{W}_{\mathbb{C}}^{SMT(\mathbb{V})(\mathbb{C})}$$

$$\Rightarrow \dim_{\mathbb{Q}} \mathbb{W}^{SMT(\mathbb{V})} = \dim_{\mathbb{C}} \mathbb{W}_{\mathbb{C}}^{SMT(\mathbb{V})(\mathbb{C})}$$

Therefore, given $SMT(\mathbb{V})(\mathbb{C})$, the computation of $\dim_{\mathbb{Q}} B^p(A)$ & $\dim_{\mathbb{Q}} End(A)_{\mathbb{Q}}$ reduces to a pure rep question.

General method: compute the branching rules for

$$SMT(\mathbb{V})(\mathbb{C}) \subseteq GL(2n;\mathbb{C})$$

e.g compute $[F_{(2n)}^{(1^{2p})}, \mathbb{C}_{trivial}]$ and $[F_{(2n)}^{(1,1)}, \mathbb{C}_{trivial}]+1$.

See [HTW05]: arXiv 0311159 for notations and formulas.

<u>In the Weil type case</u>, when $SMT(\mathbb{V})(\mathbb{C}) \cong SL(n;\mathbb{C})$,

we have
$$\mathbb{V}_{\mathbb{C}} := H^{1}(A;\mathbb{C}) \cong V \oplus V^{*},$$
where V is the standard rep of $SL(n;\mathbb{C})$. In this case,

$$\begin{aligned} H^{2p}(A;\mathbb{C}) &\cong \Lambda^{2p}\mathbb{V}_{\mathbb{C}} \\ &\cong \Lambda^{2p}(V \oplus V^{*}) \\ &\cong \bigoplus_{i+j=2p} \Lambda^{i}V \otimes \Lambda^{j}V^{*} \\ &\cong \bigoplus_{i+j=2p} \Lambda^{i}V \otimes \Lambda^{n-j}V \end{aligned}$$

$$\begin{aligned} \mathrm{End}_{\mathbb{Q}}(\mathbb{V}) \otimes_{\mathbb{Q}} \mathbb{C} &\cong \mathrm{End}_{\mathbb{C}}(\mathbb{V}_{\mathbb{C}}) \\ &\cong \mathbb{V}_{\mathbb{C}} \otimes \mathbb{V}_{\mathbb{C}}^{*} \\ &\cong (V \oplus V^{*}) \otimes (V^{*} \oplus V) \\ &\cong V \otimes V \oplus V \otimes V^{*} \oplus V^{*} \otimes V \oplus V^{*} \otimes V \end{aligned}$$

To compute $\otimes$, we need to identify $\mathrm{Rep}_{fd}(SL(n;\mathbb{C}))$ with Young diagrams, and use Littlewood-Richardson rule:

$$\begin{aligned} \emptyset &\longleftrightarrow \mathbb{C}_{\text{trivial}} \\ \square &\longleftrightarrow V \\ \begin{smallmatrix}\square\\\square\end{smallmatrix} &\longleftrightarrow \Lambda^{2}V \\ &\vdots \\ n-1 \ \begin{smallmatrix}\square\\\square\\\square\\\square\end{smallmatrix} &\longleftrightarrow \Lambda^{n-1}V \overset{SL}{\longleftrightarrow} V^{*} \\ n \ \begin{smallmatrix}\square\\\square\\\square\\\square\\\square\end{smallmatrix} &\longleftrightarrow \det V \overset{SL}{\longleftrightarrow} \mathbb{C}_{\text{trivial}} \end{aligned}$$

E.g. When $n=4$, $V = \square$, $V^* = \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix}$,

$$V \otimes V = \square \otimes \square = \begin{smallmatrix}\square\\\square\end{smallmatrix} \oplus \square\square$$

$$V \otimes V^* = \square \otimes \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} = \begin{smallmatrix}\square\\\square\\\square\\\square\end{smallmatrix} \oplus \begin{smallmatrix}\square\square\\\square\end{smallmatrix} \longrightarrow \mathbb{C}_{\text{trivial}}$$

$$V^* \otimes V^* = \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \otimes \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} = \cancel{\begin{smallmatrix}\square\\\square\\\square\\\square\\\square\\\square\end{smallmatrix}} \oplus \cancel{\begin{smallmatrix}\square\square\\\square\\\square\end{smallmatrix}} \oplus \underset{\substack{\parallel \text{ SL} \\ \begin{smallmatrix}\square\\\square\end{smallmatrix}}}{\begin{smallmatrix}\square\square\\\square\\\square\end{smallmatrix}} \oplus \begin{smallmatrix}\square\square\\\square\square\end{smallmatrix}$$

$$\Rightarrow \dim_{\mathbb{Q}} \mathrm{End}(A)_{\mathbb{Q}} = 2$$

$$H^0(A;\mathbb{C}) = \phi \otimes \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} = \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \longrightarrow \mathbb{C}_{\text{trivial}}$$

$$H^1(A;\mathbb{C}) = \phi \otimes \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \oplus \square \otimes \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix}$$

odd boxes

$$= \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \oplus \cancel{\begin{smallmatrix}\square\\\square\\\square\\\square\end{smallmatrix}} \oplus \begin{smallmatrix}\square\square\\\square\\\square\end{smallmatrix}$$

no trivial rep

$$H^2(A;\mathbb{C}) = \phi \otimes \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \oplus \square \otimes \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \oplus \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \otimes \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix}$$

$$= \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \oplus \begin{smallmatrix}\square\\\square\\\square\\\square\end{smallmatrix} \oplus \begin{smallmatrix}\square\square\\\square\\\square\end{smallmatrix} \oplus \begin{smallmatrix}\square\square\\\square\square\end{smallmatrix} \longrightarrow \mathbb{C}_{\text{trivial}}$$

$$H^4(A;\mathbb{C}) = \phi \otimes \phi \oplus \square \otimes \square \oplus \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \otimes \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \oplus \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \otimes \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \oplus \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \otimes \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix}$$

$$= \phi \oplus \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \oplus \square\square \oplus \begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} \oplus \begin{smallmatrix}\square\square\\\square\\\square\end{smallmatrix} \oplus \begin{smallmatrix}\square\square\\\square\square\end{smallmatrix} \oplus \begin{smallmatrix}\square\square\\\square\square\end{smallmatrix} \oplus \begin{smallmatrix}\square\square\\\square\square\\\square\square\end{smallmatrix}$$

Cor. For A: very general AV of Weil type, $\dim_{\mathbb{C}} A = n = 2m$,

$$\dim_{\mathbb{Q}} B^p(X) = \begin{cases} 1, & p \neq n \\ 3, & p = n \end{cases} \qquad \dim_{\mathbb{Q}} \mathrm{End}(A)_{\mathbb{Q}} = \begin{cases} 2, & n \neq 2 \\ 4, & n = 2 \end{cases}$$